

Designing and Developing a Narrative Driven Serious Game for Teaching Information Theory

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Abstract—A serious game called “Codebreakers” was developed and designed for a fourth year Telecommunications module at the University of Johannesburg. The goal of the game was to assist students to prepare for tests. The game was played for the first time in 2017 and received positive feedback. During a collaborative design session with students, educational specialists, developers and the lecturer, it emerged that students would also gain from a serious game that not only allows them to prepare for tests, but also to explore and understand the content from the start. A linear and narrative driven serious games called “EnigmAI” was designed and developed with this goal in mind. This paper explores the design and development of the new game, based on the experience gained from the previous game and from collaborative design sessions.

Keywords—Serious games, web-based learning, Information Theory

I. INTRODUCTION

Feedback from teaching and module evaluations usually include a request for more opportunities to practice relevant problems, either through more tutorials or more example questions. However, as engineering curricula are already overloaded as pointed out in [1], adding to the existing workload of students through either tutorials or assignments will only exacerbate the problem. Making more example questions and answers available, may encourage students to study questions rather than the module content. It is thus necessary to find alternative methods to assist students in preparing for assessments.

In addressing this problem directly, researchers in 2016 created a serious game called “Codebreakers”. Although there is not one commonly accepted definition for what constitutes a serious game, in this context it is accepted to be a game that is designed with not only entertainment in mind. A distinction is also made between serious games and gamification. Gamification refers to the use of game elements and game design techniques in a non-game context where serious games refer to complete games. According to [2], the following characteristics are needed for a game: Fantasy, Rules/Goals, Sensory stimuli, Challenge, Mystery and Control. Involving all stakeholders, including students and relevant experts, in the design of a game is of the utmost importance [3].

Serious games and gamification have been used in business, problem solving, creating awareness and education to

motivate, educate and engage people. An example of using serious games to aid in problem solving, is illustrated in the game “Foldit” [4]. A protein structure is posted online in the form of a puzzle, and players can interact with it. The game is made competitive with the addition of leaderboards and scores. By getting players to engage with the system, complex biochemistry problems are solved.

Creating awareness through serious games is also possible as shown in the game “Darfur is Dying” [5]. The game provides players the opportunity to experience what it is like to live in the Darfur region. Player actions involves such mundane actions as foraging for water to highlight the needs of the people of Darfur. The aim of the game is not only to create awareness but also to inspire people to take action and make a difference. Action items are embedded in the game, giving the player options of how to make a difference in the lives of the Darfur people.

Serious games also have a large footprint in education. A serious game was developed in [6] to assist the training of software managers. The game was in the form of a role playing game (RPG) where you assume the role of a project manager that needs to manage a big software project. The game also alerted the student if a wrong decision was made and explains how the situation could have been handled differently. Another RPG in the form of an Android application is proposed in [7] for STEM education. Two learning paths can be followed, one focusing on physics and one on history. A game called “Transmission” [8] is focused on the fragility of digital communication systems. Game play explores how one transmits information, both organically and digitally.

However serious games are not applicable in all situations. For example, a serious game for assessing ethics was ineffective in [9]. Games provide people with the opportunity to act out of character and try things they would not do in real life, and although this can be an enriching experience, it is not necessarily a good analogue for actual behavior.

This paper focuses on a narrative driven game designed and developed for final year Telecommunication students, focusing on the field of Information Theory. The paper is organized as follows: since the game proposed in this paper evolved from another game, that game is first described in Section II. A collaborative design session was held to evaluate the first

game. The results are discussed in Section III. The design process followed is given in Section IV. Section V contains the details of the preliminary research phase and the prototype that was developed. A second collaborative design session was held to evaluate the prototype. The results are discussed in Section VI. The design and development of the game is discussed in Section VII, before the paper is concluded in Section VIII.

II. CODEBREAKERS

A serious game called Codebreakers [10] was designed and developed for a fourth year Telecommunications module in 2016, specifically focused on Information Theory. The goal of the serious game was to give students the opportunity to practice problems in a fun and relaxed way.

The Telecommunications module is divided into two outcomes. The first outcome focuses on Information Theory and will be the focus of this paper and the relevant games. The following sections are covered in the Information Theory section of the module:

- Entropy of messages.
- Entropy of memoryless sources.
- Entropy of Markov sources.
- Source Coding - prefix codes, Huffman coding and Shannon-Fano coding.
- Error Control Coding with the focus on Hamming codes.
- Channel capacity of both discrete and continuous channels.

In addressing this need, the game was designed with the following features:

- The game consisted of different game areas, and each game area corresponded to a specific part of the work. Students thus had the option to only target a specific part of the work, if they chose to.
- The challenges are different in each section to keep the game interesting.
- The game is optional and students could decide not to play the game without any consequences.
- The challenges are randomized, thus students could play the game until they have mastered the work.
- Students' efforts are tracked and personal score boards and overall leader boards are used to motivate students.

The students played the game from February to May 2017. The success of the game was determined by looking at game session analytics and feedback from students. Students, educational experts, game developers, artists and the lecturer took part in a collaborative design session to gain further insight into how the game was perceived and to collaborate on improving the game for 2018.

III. FEEDBACK FROM THE COLLABORATIVE DESIGN SESSION

At the end of the semester, we gave students an opportunity to give feedback on their experiences. Firstly students were invited to give us feedback by using an online questionnaire. Secondly, we invited students and experts to take part in a collaborative design session.

In the questionnaire we asked students how the game could be improved. The first suggestion was that the game should be available in other formats so that students could play it on cellphones and tablets. There should also be an option to play offline. The second, and most popular suggestion, was to have additional support and feedback. Students felt that when they got stuck on a question, there should be some guidance on how to proceed. They also suggested having feedback in the form of a list of all the questions that they attempted, which ones they got incorrect, etc.

The aim of the collaborative design session was to improve Codebreakers taking the user experience into account. Students viewed the game positively and would recommend future classes to play the game. The game succeeded in its aim, which was to give students a platform to practice questions, however, students already needed a good understanding and overview of the course material before they attempted the game. The need for a game that could guide and support their learning process was identified. The possibility of a second game was suggested at the collaborative design session to address this need. This new game would be played in parallel to the existing game, since it addresses a different need. The new game would be a linear, narrative driven game.

The second part of the session was dedicated to ideas regarding the new game. It was proposed that a more narrative driven game be explored and as such, the storyline of the new game was discussed. Different options were considered including natural disasters, political/military scenarios and including trends like zombies into the game. It was also decided that the new game would be based on the Singularity, i.e. artificial intelligence taking over campus. This idea was popular since it was a topic related to the field of Electronic Engineering and students would be able to relate to it. This topic should also not make any of the students feel uncomfortable, where a topic that includes politics or the military would have to be approached with extreme caution.

It was further suggested that the game be made specifically aimed at the students at the University of Johannesburg by using familiar scenarios. For example: the robots taking over campus will be designed in the colours of a rival university. The university in the game will be based on the University of Johannesburg, students will thus immediately be engaged by the familiar surroundings, rather than using a fantasy setting.

IV. RESEARCH DESIGN AND METHODOLOGY

During the continuing development of Codebreakers, an educational design research (EDR) methodology is followed. An EDR study consists of a number of research cycles within a number of defined phases. The study is split into three main phases as proposed in [11]: the preliminary research phase, the design development and test phase, and the semi-summative evaluation phase. A detailed graphical representation of the EDR approach is given in [10]. This paper will focus on the first two stages of the new game, that started development as a consequence of feedback from the initial project. However, even though this was a new, independent game, the results

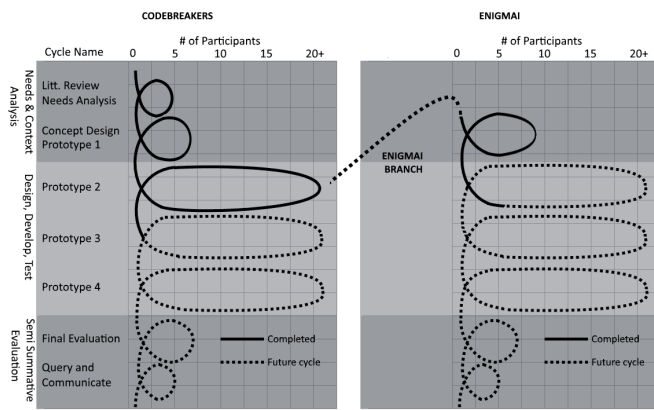


Fig. 1. Design Research Approach

and feedback from the original Codebreakers game would be taken into account and incorporated into the new game. The EDR approach as described in [10] is thus modified to reflect this, and is shown graphically in Figure 1.

The complete EDR approach for the new game is shown in Figure 2. Each one of the phases mentioned above, consists of a number of research cycles. During each stage a design is completed based on the problems encountered. The design phase consists of a number of shorter iterations known as the Agile Software Development Life Cycle.

During each cycle the design is evaluated by users, content experts and education specialists. The evaluations are done based on the guidelines given in [11] with the four criteria:

- Relevance,
- Consistency,
- Practicality, and
- Effectiveness.

The degree to which these criteria are evaluated depends on the particular design phase.

V. PRELIMINARY RESEARCH PHASE

During the preliminary research phase, a needs and context analysis is performed before a conceptual prototype is developed. Much of the need and context analysis done during the development of Codebreakers was valid for the new game as well, and could be incorporated to make this analysis more complete.

A. Needs and Context Analysis

The needs and context analysis is done following the steps proposed in [12].

- **Identifying the audience and purpose for the analysis** Fourth year students in the school of Electrical and Electronic Engineering, who are enrolled for the Telecommunications module, are the target audience. These students are familiar and comfortable with technology.
- **Describing the target population and service environment.** In Codebreakers, we were concerned whether students would have access to their own computers and

thus decided to develop an HTML5 computer game to allow students to play the game online without having to download and install any software. Students could thus play the game on any computer connected to the internet. In feedback from the students, they expressed a need to be able to download the software and play offline on devices like smart phones and tablets. Even though students preferred to play on their own devices, some of the time the game would still be played in official university spaces, like computer laboratories and study areas. This needs to be taken into account when making decision regarding music and voice overs.

- **Identifying the needs.** The purpose of the game is to assist students in exploring and understanding the course material more holistically. By playing the game, they would systematically be introduced to different aspects of Information Theory. Puzzles and challenges would be incorporated in such a way to encourage students firstly to think about information theoretic problems, before continuing in solving them. The game would thus encourage learning through exploration.

The need identified in the Codebreakers questionnaire of more support would also be addressed. Students would have the opportunity to refresh concepts that were explained in class at specific points in the game, by watching short videos explaining important concepts. This would not only help students if they get stuck, but also enable students to progress through the game at their own speed. Multiple other options would be proposed to students when they are struggling with problems. Except for the videos, students would be referred to the relevant module notes, the textbook and lastly to contact the lecturer.

- **Assessing the needs.** Possible methods of meeting the identified goal were discussed in the collaborative design session. It was decided that a narrative based serious game would be applicable as the goal is to guide students through all of the coursework. In Codebreakers, students could target a specific part of the game as needed. In the new game, problems would be solved sequentially. Students would thus be forced to master a specific problem before progressing to the next problem. If a student could not solve a problem, different interventions would be made available: the student will be directed to the relevant lecture notes, short videos or the lecturer.

The initial needs identified for Codebreakers were still relevant. Codebreakers would thus be available parallel to the new game. Students could play the new game to familiarize themselves with the new concepts and then use Codebreakers to exercise the new concepts.

- **Communicating and collating the results.** As an EDR approach is followed, communication of the results is constant with all stakeholders involved in the development process directly. Results gained from evaluating the effectiveness in addressing the needs of students are a key feedback mechanism. Communication of the results

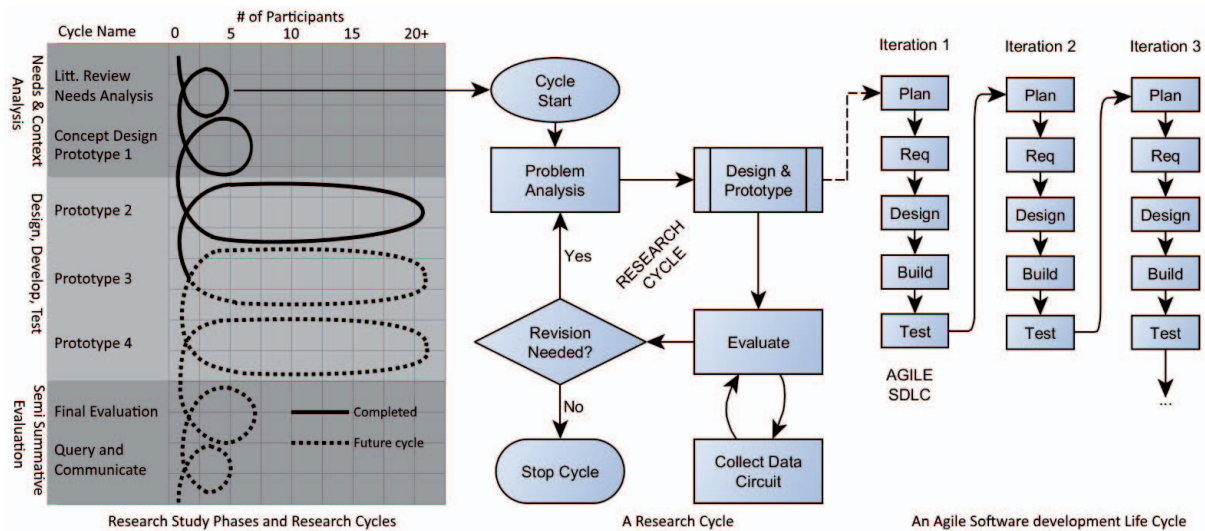


Fig. 2. Complete EDR approach for EnigmAI

to the wider community is done through more formal publications.

B. Conceptual Prototype

After the needs and context analysis, a conceptual prototype was developed. The prototype was separated into two parts. One part focused on the narrative and how problems and challenges would be incorporated into the game. The second part focused on the environment, graphics and game mechanics which would be used. Once the two different parts were evaluated, it would be easy to expand the concepts and combine them into a complete game.

For the first part, an open-source tool called Twine was used to create an interactive fiction game. This enabled the lecturer to design and test the flow of the game without having to take game design elements, for example graphics, into account. The integration of the different problems into the game narrative could be tested with minimal coding effort. The game could be published as HTML so it could be easily tested online or locally regardless of the platforms used by stakeholders for testing and quality control. Development in Twine requires minimal programming knowledge and could also be used by lecturers not familiar with game programming. Feedback and suggestions relating to the narrative raised during the collaborative design sessions are easily incorporated back into Twine.

Twine was chosen as the tool of choice for exploring the narrative element as there exists a workflow for using the interactive fiction created in Twine directly in Unity 5.0 by using the Cradle Plugin. This allows the published narrative to be used as a state machine to trigger events and custom scripts in Unity effectively shifting the development burden out of the Unity Framework. The Unity Framework can also export games to a number of platforms, meeting the requirement from the student feedback.

In Figure 3, an example is shown of the first part of the

narrative, created in Twine and played in a browser. At the end of the page, the player has two options, "Phone" or "Search." Each of these options takes the player to a different part in the narrative. However, once the narrative is integrated into the game, these options would not be available in text but through interacting with game objects. For example: rather than choosing "Phone", the player would see the phone and pick it up to make a call.

The second part of the prototype focused on the look and feel of the game. A number of models were created to show what the game world would look like. Graphics from Codebreakers are also included to show how the new game's graphics differ from the old game. Figure 4 shows the player character from Codebreakers, as well as what the office looked like. Figure 5 shows what an office looks like in the new game. In modeling the game world based on the University of Johannesburg, the new game would be in a familiar, real life environment, in contrast to the fantasy world of Codebreakers.

The new game would also be from a first person perspective, assisting in making the player feel like they are in the game themselves. The design for the robotic protagonist is shown in Figure 6. The robot can be compared to the Codebreakers character shown in Figure 4.

It was difficult in Codebreakers to get input from the user in the form of text. In the new game, one can easily use components like the keypad shown in Figure 7. The difference between the switches used in the two games are shown in Figures 8 and 9.

The proposed name of the new name is EnigmAI. The reference to the Enigma machine is also a reference to Claude E. Shannon, the father of Information Theory, who was also involved in cracking the code of the Enigma machine. The word "enigma" also means puzzle or mystery. The A.I. stands for artificial intelligence, referring to the Singularity taking place in the game. The logos of Codebreakers and EnigmAI are given in Figures 10 and 11 respectively.

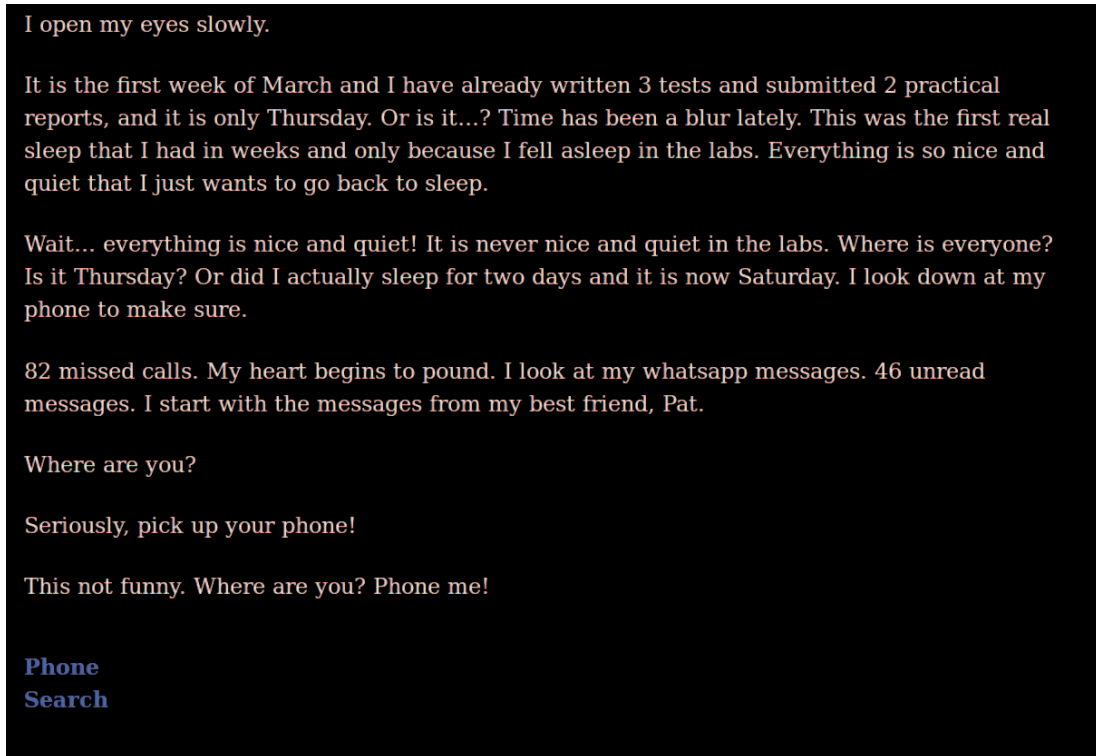


Fig. 3. Example from Twine



Fig. 4. Codebreakers: person and world.

VI. SECOND COLLABORATIVE DESIGN SESSION

After the prototype was created, a second collaborative design session was held. The group consisted of the module lecturer, the lead game developer, an educational specialist and a student who played the first game. Everyone was also part of the first collaborative design session. The goal of the session



Fig. 5. Office

was to show everyone that was part of the previous design session, but not part of the development team, how the ideas in the first session were implemented and to get feedback on whether the new game was what they had in mind.

A number of questions were prepared before the session to guide the discussion. The questions are divided in four main sections:

- characters,
- the game,
- the concept, and
- the overworld elements.

A summary of each of these sections is given in the follow-



Fig. 6. Robot



Fig. 7. Keypad

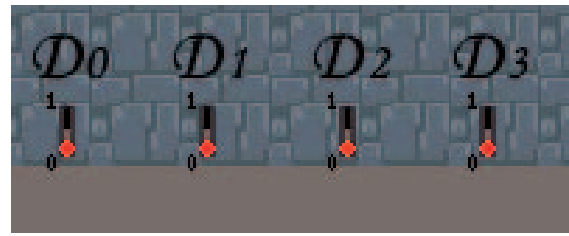


Fig. 8. Codebreakers - switches



Fig. 9. Switches

ing subsections, focusing on the most important feedback.

A. Characters

During our initial collaborative design session on Codebreakers, the feedback on the characters was that the students did not care who the characters were or what they looked like. The focus was rather on attempting as many problems as possible. However, when a narrative driven game was proposed, students stated that this type of game changes how they feel about the game characters, since there is a feeling of the player being in the game personally. The characters and environments would impact their experience when playing a narrative driven game.

The participants liked the idea of the game being a first person game, thus creating the feeling that you are in the game.



Fig. 10. Codebreakers logo

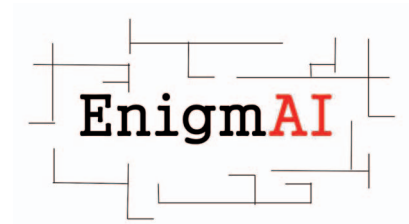


Fig. 11. EnigmAI logo

They also liked the prototype robot character, shown in Figure 6. They proposed that, even though the robot has achieved consciousness, it should still have robotic characteristics, for example jerky movements and a robotic voice. The robot should rather be conniving, than obviously evil. It should try to be your friend and gain your trust, even though being your nemesis. The robot should also be able to manipulate the environment to confuse the player.

In Codebreakers, Dr. H. is the villain. Dr. H. is a reference to the module lecturer. The designers of Codebreakers thought it was a good idea to make the lecturer the student's nemesis. However, in the initial collaborative design session, students said that, although they enjoyed the character, they did not like the idea that the lecturer was the villain. They would prefer it if Dr. H. is in the new game but as a support character. It was proposed that the player in EnigmAI should be in contact with Dr. H. telephonically. She will give advice, but at strategic points the connection will break up and the student will be left to solve the challenge.

B. The game

The participants like the familiarity of the game. It is set in the Department of Electrical and Electronic Engineering and the environment is created to look similar to the real-life environment. They proposed that objects can have fake keys, created by the robot, and the player should have to figure out which one is correct. For example, in the office in Figure 5, one can see a book on the desk which contains information that the player can use. It was proposed that there should rather be two books that look similar with the one authored by Claude E. Shannon and the other by Claude E. Sharron. The player should then be able to make a decision about which book is the real key and which one is the fake one.

EnigmAI also has a darker atmosphere and feel to it compared to Codebreakers. The participants approved of the use of the Horror Adventure genre. The overall feeling was that the more adult theme would encourage students to firstly play the game, and secondly have a more enjoyable experience. The game will be set at night and the user would move around by using a torch, as is shown in Figure 5.

Simple and clean menu systems are also preferred by the participants, and less animation. For example, the fact that everything is so quiet contributes to the atmosphere of the game. If there is movement on the screen, it would immediately draw the attention of the player and could be used in the overall storyline to introduce jump scares. The participants liked the horror theme, but would prefer a creepy atmosphere than any actual violence being introduced.

C. Concept

Everyone agreed that an adventure game was a good choice as a genre for a serious game. Combining that with the horror theme, will encourage engagement. Everyone was of the opinion that the game would be successful in assisting with discovering and learning new concepts.

The student noted that it was important to make sure that, even though the player enjoys the game, they are also learning something all the time. It would be negative for the students if the puzzles were spaced too far apart in the game world. Even if students are enjoying the game, they do not want to feel like they are wasting their time.

The initial idea was that the game would consist of a number of predetermined challenges and puzzles, almost like a tutorial. Codebreakers will still run in parallel to EnigmAI and provide students with a large number of randomized questions. However, the participants felt that some randomization was necessary in EnigmAI. The pool of questions could be much smaller than in Codebreakers, but it should not be possible to just get an answer from a friend.

D. Overworld elements

Being at the engineering department at night is a scary experience in itself. A big contributor to that feeling is the fact that it is completely quiet, in contrast to the day time when there is always lots of activity. It was thus decided not to include music in the game. Another advantage of this is that students could play the game in computer laboratoria without bothering other students.

Participants liked the overall colour palette, specifically the darker colours which contributed to the atmosphere. They felt that the speed with which the characters moved was good as long as it was controllable.

VII. DEVELOPMENT PHASE

Taking the feedback from the collaborative design session into account, the first version of the full game was developed. The game is described in the following subsections, following the attributes of games as described in [3].

A. Backstory and storyline

The game starts with the player waking up in one of the laboratories. It was a busy week, as always in engineering, and at some point the student just fell asleep. However, it is now dark and the player is all alone. The player realizes through interaction with different game objects, that the campus has been evacuated. The player makes contact with Dr. H., the Telecommunications lecturer. Since the player is the only person left on campus, it is up to the player to disable the robot(s) before they cause greater damage.

The player needs to move to certain locations. Challenges and puzzles are distributed through the game. These puzzles and challenges are Information Theory specific, but also embedded in the game to form part of the storyline.

B. Immersive Graphical environment

Graphics are, where possible, sourced from free assets available on the internet. A small percentage of generic graphics are purchased online. A graphic design student is also used to create the most important, non-generic objects specific to the game.

C. Interactivity

A 3-dimensional graphical world is created based on the University of Johannesburg Department of Electrical and Electronic Engineering. The player can walk around freely and interact with elements in the environment, for example picking-up a book and reading what is inside. However, the game follows a specific narrative and the user can only proceed through the game by attempting challenges in a specific, pre-determined order. This is to assist the student in exploring the module's material in a sensible way. The order will also mirror the order in which concepts are discussed in class.

D. Challenge or Competition

Players time in the game will be recorded and displayed on an Overall Leaderboard. However, smaller achievements will also be tracked and used to encourage players. For example: the number of times a student attempts a puzzle is tracked as well as the time spent on individual puzzles. Feedback will also be given on how they are faring against their peers. For example: if you reach a challenge, the number of students who have attempted the challenge is displayed along with the success rate.

E. Risk and consequences

There is no risk or consequence involved for students by playing, or not playing, the game. Playing the game is optional and will not directly affect the marks of the students. This was also discussed by the participants at the collaborative design session, and everyone agreed that the game would be enticing enough to encourage students to play it, without being forced.

VIII. CONCLUSION

After the success of Codebreakers, a game to assist students in preparation for assessments in Information Theory, a need was identified to create a serious game that could guide students in exploring the module content. The new game is primarily a narrative driven game, where students need to master a specific part of the work before proceeding with the next part of the game.

A text based prototype was created in Twine and tested by students, developers and content experts. A collaborative design session approach was used to get feedback on the

prototype and design the next iteration of the game. In the next development phase, the text based game will be ported to the game engine Unity, to create a 3-dimensional game, using the created Twine Interactive Fiction as the base through the use of Cradle.

ACKNOWLEDGMENT

The authors would like to sincerely thank everyone who contributed to this game by participating in the Collaborative Design workshops, by filling in questionnaires or by playing the game. A special tanks to Mr. L. Bunt, for his design approach for the collaborative design session. Lastly, thank you to the University of Johannesburg's Teaching and Innovation Fund, for supporting this project.

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